THE TWO-THIRDS THEOREM:
A MATRIX PROOF

Abstract. An unconditional theorem proves, without assuming the Riemann hypothesis, that at least two thirds of the nontrivial zeros of the Riemann zeta function are simple and on the critical line, and that at least five sixths are distinct. This note explains that proof; it claims no new zeta theorem. The key step uses three facts about a real symmetric matrix: the sum of its eigenvalues, the sum of their squares, and the number that are positive. Those facts give both zero counts. The same proof gives the dyadic and cumulative forms, the Montgomery–Taylor constants, and the result for each fixed primitive Dirichlet character. The appendix states which claims occur in the accompanying Lean files and which build claims were not independently rerun.

1. The theorem and its structure

Write a nontrivial zero of ζ as $\rho = \beta + i\gamma$, and let m_ρ denote its multiplicity. For $0 \le T_1 < T_2$, let

$$\begin{aligned} N(T_1,T_2) &= \sum_{T_1<\gamma\le T_2} m_\rho, \\ N_d(T_1,T_2) &= \#\{\rho : T_1 < \gamma \le T_2\}, \\ N_0^*(T_1,T_2) &= \#\{\rho : T_1 < \gamma \le T_2,\ \beta = \tfrac12\}, \\ N_0(T_1,T_2) &= \sum_{\substack{T_1<\gamma\le T_2 \\ \beta=1/2}} m_\rho, \\ N_0^s(T_1,T_2) &= \#\{\rho : T_1 < \gamma \le T_2,\ \beta = \tfrac12,\ m_\rho = 1\}. \end{aligned}$$

Write also $N^s(T_1,T_2)$ for the simple zeros without the critical-line restriction. Thus N and N_0 count with multiplicity, while the other functions count distinct points. We call a zero on-line when $\beta = 1/2$ and off-line otherwise. The theorem below is proved in [8]; we explain its proof.

Theorem 1.1. *As $T \to \infty$,*

$$N_0^s(T,2T) \ge \left(\frac{2}{3} - o(1)\right) N(T,2T), \tag{1}$$

$$N_d(T,2T) \ge \left(\frac{5}{6} - o(1)\right) N(T,2T). \tag{2}$$

For the Montgomery-Taylor window, the constants are respectively

$$2 - c_{\mathrm{MT}}^{-1}, \qquad \frac{3 - c_{\mathrm{MT}}^{-1}}{2}, \qquad c_{\mathrm{MT}}^{-1} = \frac12 + \frac{1}{\sqrt2}\cot\left(\frac{1}{\sqrt2}\right). \tag{3}$$

The same statements hold on $(0,T)$, and hold verbatim for the L-function of every fixed primitive Dirichlet character. Moreover,

$$N_0^*(T,2T),\ N_0(T,2T),\ N^s(T,2T) \ge \left(\frac{2}{3} - o(1)\right) N(T,2T), \tag{4}$$

Date: August 16, 2026.

2020 *Mathematics Subject Classification.* Primary 11M06; Secondary 11M26, 15A42.

Key words and phrases. Riemann zeta function, simple zeros, critical line, Weil form, Hermitian inertia.

and the same a fortiori bounds hold cumulatively. For every fixed window (and fixed character in the Dirichlet case), the error in these proportions may be taken to be $O(\log\log T/\log T)$*.*

Here $o(1)$ means that for each $\varepsilon > 0$ there is T_0 such that the inequality with $o(1)$ replaced by ε holds for every $T \geq T_0$. No Riemann hypothesis, zero-density estimate, zero-free region, or mollifier is assumed.

The proof makes four main moves.

First, use the functional equation to mirror each off-line zero across the critical line. The pair can contribute at most one positive and one negative direction. Thus we count its positive directions instead of asking each term to be positive.

Second, do not find every eigenvalue of the large matrix. Keep only its trace, its squared Hilbert–Schmidt norm, and the number of positive eigenvalues from the nonsimple or off-line part. The trace is the sum of the eigenvalues, and the squared norm is the sum of their squares.

Third, put those three quantities into one rank–trace inequality. The same calculation gives both zero counts.

Finally, split the prime-side second moment into $n = m$ and $n \neq m$. The diagonal gives the main term, and the Montgomery–Vaughan Hilbert inequality bounds the rest. In symbols,

$$\boxed{\text{explicit formula and sampling}} \longrightarrow \boxed{\operatorname{tr} G \text{ and } \|G\|_{\mathrm{HS}}^2} \longrightarrow \boxed{\text{rank–trace inequality}} \longrightarrow \boxed{N_0^s,\ N_d}.$$

The first arrow is analytic number theory. The rest is finite linear algebra and exact counting.

2. The matrix inequality

A real symmetric matrix satisfies $A^{\mathsf{T}} = A$; a Hermitian matrix is its complex analogue, satisfying $A^* = A$. Both have real eigenvalues. We write $A \succeq 0$ when all those eigenvalues are nonnegative, $\operatorname{rank} A$ for the number of nonzero eigenvalues, and $n_+(A)$ for the number of strictly positive eigenvalues, always counted with multiplicity. Moreover,

$$\operatorname{tr} A = \sum_i \lambda_i(A), \qquad \|A\|_{\mathrm{HS}}^2 = \operatorname{tr}(A^2) = \sum_i \lambda_i(A)^2.$$

For a column vector v, the matrix vv^{T} has rank at most one; when v is real it is positive semidefinite. These elementary facts are the entire matrix vocabulary used below.

The rank–trace inequality used below is proved in [8, Lemma 3.2]; we include its proof.

Lemma 2.1 (rank-trace inequality)**.** *Let P, Q be Hermitian $d \times d$ matrices. Suppose that $P \succeq 0$, $\operatorname{rank} P \leq r$, and $n_+(Q) \leq b$. Then*

$$r \geq 2\operatorname{tr} P + 4\operatorname{tr} Q - 4b - \|P + Q\|_{\mathrm{HS}}^2. \tag{5}$$

The coefficients are sharp: equality holds for $P = \Pi_1$ and $Q = 2\Pi_2$ when Π_1, Π_2 are orthogonal projections of ranks r, b with $\Pi_1\Pi_2 = 0$.

Proof. Write $Q = Q_+ - Q_-$ with $Q_\pm \succeq 0$, $Q_+Q_- = 0$, and $\operatorname{rank} Q_+ \leq b$. If p_i, n_i are the decreasing eigenvalue lists of P, Q_-, then

$$\|P + Q\|_{\mathrm{HS}}^2 = (\|P\|_{\mathrm{HS}}^2 - 2\operatorname{tr}(PQ_-) + \|Q_-\|_{\mathrm{HS}}^2) + 2\operatorname{tr}(PQ_+) + \|Q_+\|_{\mathrm{HS}}^2.$$

Here the mixed term $\operatorname{tr}(Q_-Q_+)$ vanishes because $Q_-Q_+ = 0$. Von Neumann's trace inequality says $\operatorname{tr}(PQ_-) \leq \sum_i p_i n_i$, and therefore the parenthesized expression is at least $\sum_i (p_i - n_i)^2$. For each $i \leq r$ the exact scalar identity

$$(p_i - n_i)^2 - (2p_i - 1 - 4n_i) = (p_i - n_i - 1)^2 + 2n_i \geq 0$$

holds. For $i > r$ one has $p_i = 0$ and $(p_i - n_i)^2 = n_i^2 \geq 0$. Hence

$$\sum_i (p_i - n_i)^2 \geq 2\sum_{i\leq r} p_i - r - 4\sum_{i\leq r} n_i \geq 2\operatorname{tr} P - r - 4\operatorname{tr} Q_-.$$

Positivity gives $\operatorname{tr}(PQ_+) \geq 0$. On each positive eigenvalue q of Q_+, the elementary square $(q-2)^2 \geq 0$ is exactly $q^2 \geq 4q - 4$. Since there are at most b such eigenvalues, $\|Q_+\|_{\mathrm{HS}}^2 \geq 4\operatorname{tr} Q_+ - 4b$. Adding the two bounds and using $\operatorname{tr} Q = \operatorname{tr} Q_+ - \operatorname{tr} Q_-$ proves (5). The stated projections make each scalar inequality an equality. □

Remark 2.2 (other coefficients). For $c \geq 0$, the homogeneous coefficient version is

$$\frac{c^2}{4}\operatorname{rank} P \geq c\operatorname{tr} P + 2c\operatorname{tr} Q - c^2 b - \|P + Q\|_{\mathrm{HS}}^2.$$

The orthogonal-projection equality case is $P = (c/2)\Pi_1$ and $Q = c\Pi_2$. At $c = 2$, the value used here, this is exactly $P = \Pi_1$ and $Q = 2\Pi_2$.

The next proposition states the zero-counting step used below [8, Sections 3–6].

Proposition 2.3 (matrix zero-counting bound)**.** *Let $M \to \infty$. For each M, suppose that a real symmetric matrix G comes with nonnegative integers s_1, s_2, p and a decomposition $G = P_1 + Q'$ such that*

$$P_1 \succeq 0, \qquad \operatorname{rank} P_1 \leq s_1, \qquad \operatorname{tr} P_1 \leq s_1, \qquad n_+(Q') \leq s_2 + p, \tag{6}$$

$$s_1 + 2s_2 + 2p \leq M + o(M), \tag{7}$$

$$\operatorname{tr} G = M + o(M), \qquad \|G\|_{\mathrm{HS}}^2 = (R + o(1))M. \tag{8}$$

Then

$$s_1 \geq (2 - R - o(1))M, \tag{9}$$

$$s_1 + s_2 + p \geq \frac{3 - R - o(1)}{2}M. \tag{10}$$

Proof. Apply Lemma 2.1 to (P_1, Q'). Since $\operatorname{tr} Q' = \operatorname{tr} G - \operatorname{tr} P_1$,

$$\operatorname{rank} P_1 \geq 4\operatorname{tr} G - 2\operatorname{tr} P_1 - 4(s_2 + p) - \|G\|_{\mathrm{HS}}^2. \tag{11}$$

Using (8), rewrite this as

$$\operatorname{rank} P_1 \geq (4 - R - o(1))M - 2\big[\operatorname{tr} P_1 + 2(s_2 + p)\big].$$

The bracket is at most $M + o(M)$ by (7) and $\operatorname{tr} P_1 \leq s_1$. Thus $\operatorname{rank} P_1 \geq (2 - R - o(1))M$, and $\operatorname{rank} P_1 \leq s_1$ gives the first claim.

For the second assertion, combine (11) with $\operatorname{rank} P_1 \leq s_1$ and $\operatorname{tr} P_1 \leq s_1$ to obtain

$$3s_1 + 4(s_2 + p) \geq (4 - R - o(1))M.$$

Subtracting $s_1 + 2(s_2 + p) \leq M + o(M)$ from this inequality gives

$$2s_1 + 2(s_2 + p) \geq (3 - R - o(1))M,$$

which is the second claim after division by two. □

This matters for two reasons. We charge an off-line pair only for its positive index, and one calculation gives both counts. At $R = 4/3$ the outputs are $2/3$ and $5/6$.

Here is what the variables count. The third column is the minimum charge to the multiplicity count $N(I')$.

Zero class	Number	Multiplicity charge	Matrix role
Simple, on the line	s_1	1	positive rank-one term in P_1
Multiple, on the line	s_2	2	at most one positive direction in Q'
Off-line reflected pair	p	2	at most one positive direction in Q'

Thus $s_1 + 2s_2 + 2p \le M + o(M)$ tracks multiplicity. The identities $\operatorname{tr} G = M + o(M)$ and $\|G\|_{\mathrm{HS}}^2 = (R + o(1))M$ give the first and second moments. Proposition 2.3 turns these three facts into the two zero counts.

3. How zeta zeros produce the matrix data

We now show how zeta zeros give the data used in Proposition 2.3. This follows [8]; the source contains the full endpoint and error details.

Let $L = \log(T/2\pi)$, $X = e^L = T/(2\pi)$, and distinguish the analytic interval $I_{\mathrm{an}} = [T, 2T)$ from the theorem interval $I_{\mathrm{th}} = (T, 2T]$. Their endpoint counts differ by $O(\log T) = o(N(T, 2T))$. Choose an even function $\psi \in C^2([-1/2, 1/2])$ that is strictly positive on the closed interval, and fix a nondecreasing $\vartheta \in C^\infty(\mathbb{R})$ equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$. Set

$$\varphi(u) = \vartheta(L/2 + u)\vartheta(L/2 - u)\psi(u/L)^{1/2}. \tag{12}$$

Then φ is real, even, and C^2, is supported on $[-L/2, L/2]$, and satisfies $\varphi(u)^2 = \psi(u/L)$ away from fixed-width endpoint transition intervals. Moreover,

$$a := L^{-1}\|\varphi\|_2^2 = \int_{-1/2}^{1/2} \psi(u)\,du + O_\psi(L^{-1}), \quad \|\varphi\|_1 \ll_\psi L, \quad \|\varphi'\|_1 + \|\varphi''\|_1 \ll_\psi 1.$$

Put

$$\alpha_k = T + \frac{2\pi k}{L}, \qquad d = \left\lfloor \frac{LT}{2\pi} \right\rfloor, \qquad v_\rho = \big(\widehat{\varphi}(\gamma_\rho - \alpha_k)\big)_{0 \le k < d}.$$

Our Fourier-transform convention is $\widehat{f}(\xi) = \int_{\mathbb{R}} f(u)e^{-iu\xi}\,du$, and $(f \star g)(y) = \int_{\mathbb{R}} f(u)g(y-u)\,du$. Here $\gamma_\rho = (\rho - 1/2)/i$, which is real precisely on the critical line. We use the following normalized finite matrix from Weil's Hermitian form:

$$G = \frac{1}{aL^2} \sum_{\Re\gamma_\rho \in I'} m_\rho v_\rho v_\rho^{\mathsf{T}}, \qquad I' = [T - \sqrt{T}, 2T + \sqrt{T}). \tag{13}$$

Let E denote the same normalized sum over zeros with $\Re\gamma_\rho \notin I'$. Thus $G + E$ is the full sampled Weil matrix, while G is the finite matrix used for counting. The transpose, rather than conjugate transpose, is forced by the bilinear form in the explicit formula. The functional equation pairs ρ with $1 - \overline{\rho}$ and makes (13) real symmetric.

3.1. **The exact sampling identity.** The spacing $2\pi/L$ is chosen so that Poisson summation gives the following identity for all $z, z' \in \mathbb{C}$:

$$\sum_{k \in \mathbb{Z}} \widehat{\varphi}(z - \alpha_k)\widehat{\varphi}(z' - \alpha_k) = L\widehat{\varphi^2}(z - z'). \tag{14}$$

Indeed, the relevant convolution is supported in $[-L, L]$ and vanishes at its endpoints, while the dual sampling lattice is $L\mathbb{Z}$; consequently only the zero dual frequency survives. For real $z = z'$, (14) gives

$$\|v_\rho\|_2^2 \le L\|\varphi\|_2^2 = aL^2. \tag{15}$$

3.2. **Signs from on-line and off-line zeros.** On-line zeros have $v_\rho \in \mathbb{R}^d$ and contribute positive rank-one forms. Instead of pairing ρ with $\overline{\rho}$, which reverses its ordinate, the functional equation pairs it with $1-\overline{\rho}$; this preserves positive height and gives $\gamma_{1-\overline{\rho}} = \overline{\gamma_\rho}$. If an off-line pair is written $v_\rho = u + iw$, then its combined contribution is

$$m_\rho(v_\rho v_\rho^\mathsf{T} + \overline{v}_\rho \overline{v}_\rho^\mathsf{T}) = 2m_\rho(uu^\mathsf{T} - ww^\mathsf{T}). \tag{16}$$

Equation (16) shows that each off-line pair comes from the matrix $\operatorname{diag}(1,-1)$, which has one positive eigenvalue. After the positive multiplicity and normalization factors are absorbed into A, all nonsimple on-line and off-line contributions have the form

$$Q' = A^\mathsf{T} \left(\bigoplus_{j=1}^{s_2} [1] \;\oplus \bigoplus_{j=1}^{p} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right) A,$$

For any real A, the matrix $A^\mathsf{T} D A$ has no more positive eigenvalues than D. The displayed direct sum has $s_2 + p$ positive eigenvalues, so $n_+(Q') \le s_2 + p$. This does not require the summands to be orthogonal.

Let s_1 be the number of simple on-line points in I', s_2 the number of distinct multiple on-line points, and p the number of off-line functional- equation pairs. Put P_1 equal to the sum in (13) over the simple on-line zeros and $Q' = G - P_1$. Equations (15) and (16) give

$$P_1 \succeq 0, \qquad \operatorname{rank} P_1 \le s_1, \qquad \operatorname{tr} P_1 \le s_1, \qquad n_+(Q') \le s_2 + p, \qquad s_1 + 2s_2 + 2p \le N(I'). \tag{17}$$

The last inequality is deliberately weak: a multiple point has multiplicity at least two, and each off-line pair contributes at least two zeros.

The wider interval I' leaves room at both ends for the Fourier tail to decay before the zero sum is cut off. The local Riemann–von Mangoldt bound gives

$$N(I' \setminus I_{\mathrm{th}}) = O(\sqrt{T} \log T) = o(N(T, 2T)).$$

Two integrations by parts give $|\widehat{\varphi}(r + iy)| \ll e^{L|y|/2} \min(L, |r|^{-1}, |r|^{-2})$. Together with $N(t, t+1) \ll \log(|t| + 3)$ and the buffer $\sqrt{T}$, this yields

$$\|E\|_1 \ll T^{-1/2}, \qquad \operatorname{tr} G = N(I') + O(\sqrt{T} L^2) = N(T, 2T) + o(N(T, 2T)). \tag{18}$$

This gives the first moment and the needed multiplicity bound.

4. The Hilbert-Schmidt moment

The second moment is where the explicit formula enters. In the normalization above, the analytic statement needed by the transfer principle is the following.

Proposition 4.1 (prime-side moment)**.** *For the window ψ and matrix G above,*

$$\|G\|_{\mathrm{HS}}^2 = \big(R(\psi) + o(1)\big) N(T, 2T), \tag{19}$$

where

$$R(\psi) = \frac{\displaystyle\int_{-1/2}^{1/2} \psi(u)^2 \, du + \int_{-1/2}^{1/2}\int_{-1/2}^{1/2} |u - v| \psi(u) \psi(v) \, du \, dv}{\left(\displaystyle\int_{-1/2}^{1/2} \psi(u) \, du \right)^2}. \tag{20}$$

To prove the proposition, define

$$\mu(\tau) = \frac{1}{2\pi}\Re\frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{i\tau}{2}\right) - \frac{\log\pi}{2\pi},$$

$$\Pi_X(\tau) = \frac{1}{\pi}\Re\frac{X^{1/2+i\tau}}{1/2+i\tau}, \qquad P_X(\tau) = -\frac{1}{\pi}\sum_{n\le X}\frac{\Lambda(n)}{\sqrt{n}}\cos(\tau\log n),$$

$$\nu_X(\tau) = \mu(\tau) + \Pi_X(\tau) + P_X(\tau). \tag{21}$$

The three terms are, respectively, the archimedean term, the two pole terms combined into one real function, and the prime sum. For every even $F \in C_c^2(\mathbb{R})$ supported in $[-\log X, \log X]$, Weil's explicit formula gives

$$\sum_\rho m_\rho \widehat{F}(\gamma_\rho) = \int_{\mathbb{R}} \widehat{F}(\tau)\nu_X(\tau)\,d\tau. \tag{22}$$

Apply this entry by entry to the products of modulated copies of φ. It gives

$$(G+E)_{kk'} = \frac{1}{aL^2}\int_{\mathbb{R}} \widehat{\varphi}(\tau-\alpha_k)\widehat{\varphi}(\tau-\alpha_{k'})\nu_X(\tau)\,d\tau.$$

Now put

$$K(\tau,\tau') = \sum_{0\le k<d}\widehat{\varphi}(\tau-\alpha_k)\widehat{\varphi}(\tau'-\alpha_k).$$

Expanding $\|G+E\|_{\mathrm{HS}}^2 = \mathrm{tr}((G+E)^2)$ first gives the exact identity

$$(aL^2)^2\|G+E\|_{\mathrm{HS}}^2 = \iint_{\mathbb{R}^2} K(\tau,\tau')^2\nu_X(\tau)\nu_X(\tau')\,d\tau\,d\tau'.$$

The finite grid means that K is not itself the full Poisson sum. Write

$$K(\tau,\tau') = L\widehat{\varphi^2}(\tau-\tau') - K_{\mathrm{out}}(\tau,\tau'),$$

where K_{out} is the sum over $k \notin [0,d)$. The source estimates K_{out} using the decay of $\widehat{\varphi}$ and controls the signed weight ν_X by absolute-value bounds; no positivity of ν_X is assumed. Those estimates replace K by the full Poisson sum, restrict $\mathbb{R}^2$ to $I_{\mathrm{an}} \times I_{\mathrm{an}}$, and remove E, with normalized error $O_\psi(N\log L/L^2) = o(N)$. What remains is

$$\frac{1}{a^2L^2}\iint_{I_{\mathrm{an}}\times I_{\mathrm{an}}} \widehat{\varphi^2}(\tau-\tau')^2\nu_X(\tau)\nu_X(\tau')\,d\tau\,d\tau'. \tag{23}$$

The archimedean diagonal uses $\mu(\tau) = (2\pi)^{-1}\log(\tau/2\pi) + O(\tau^{-2})$ and contributes

$$\frac{TL^3}{2\pi}\int_{-1/2}^{1/2}\psi(u)^2\,du + O(TL^2). \tag{24}$$

For the prime-prime term, write the two copies of P_X as double sums over n and m. Instead of estimating all pairs together, split them into $n = m$ and $n \ne m$. Fourier inversion changes the diagonal into the autocorrelation

$$g(y) = (\varphi^2 \star \varphi^2)(y) = \int_{\mathbb{R}} \varphi(u)^2\varphi(y-u)^2\,du$$

and gives

$$\frac{T}{\pi}\sum_{n\le X}\frac{\Lambda(n)^2}{n}g(\log n). \tag{25}$$

Partial summation from $\sum_{n\le x}\Lambda(n)^2/n = \frac{1}{2}\log^2 x + O(\log x)$ turns (25) into

$$\frac{TL^3}{2\pi}\int_{-1/2}^{1/2}\int_{-1/2}^{1/2}|u-v|\psi(u)\psi(v)\,du\,dv + O(TL^2). \tag{26}$$

The off-diagonal $n \neq m$ is $O(L^2 X) = o(TL^3)$ by the Montgomery–Vaughan Hilbert inequality applied to the distinct frequencies $\log n$ [14]; the pole and mixed terms are $O(L^2\sqrt{X})$. Equations (24) and (26), divided by $a^2L^2 = L^2(\int \psi)^2 + O(L)$, prove (19). These prime-side estimates come from work of Montgomery, Aryan, and Baluyot–Goldston–Suriajaya–Turnage-Butterbaugh [12, 3, 4].

5. The constants and the final count

For the flat window $\psi_0 = \mathbf{1}_{[-1/2,1/2]}$,

$$\int \psi_0 = 1, \qquad \int \psi_0^2 = 1, \qquad \iint |u-v|\psi_0(u)\psi_0(v)\,du\,dv = \frac{1}{3},$$

so $R(\psi_0) = 4/3$. For $\psi_{\mathrm{MT}}(u) = \cos(\sqrt{2}u)\mathbf{1}_{[-1/2,1/2]}(u)$, put

$$H(u) = \psi_{\mathrm{MT}}(u) + \int_{-1/2}^{1/2} |u-v|\psi_{\mathrm{MT}}(v)\,dv.$$

On the interior, $H'' = \psi_{\mathrm{MT}}'' + 2\psi_{\mathrm{MT}} = 0$. Evenness makes H constant. Direct integration gives

$$H(0) = \cos\left(\frac{1}{\sqrt{2}}\right) + \frac{1}{\sqrt{2}}\sin\left(\frac{1}{\sqrt{2}}\right), \qquad \int_{-1/2}^{1/2} \psi_{\mathrm{MT}}(u)\,du = \sqrt{2}\sin\left(\frac{1}{\sqrt{2}}\right).$$

Integrating $H\psi_{\mathrm{MT}}$ therefore gives

$$R(\psi_{\mathrm{MT}}) = \frac{1}{2} + \frac{1}{\sqrt{2}}\cot\left(\frac{1}{\sqrt{2}}\right) = c_{\mathrm{MT}}^{-1}. \tag{27}$$

The optimality of this window within the corresponding variational class is the Hilbert-space extremal problem solved in [6]. The displayed sharp windows are used through the C^2 endpoint smoothing in (12); the transition intervals affect only the stated lower-order terms.

Take $M = N(T, 2T)$. Equations (17), (18), and Proposition 4.1 give the hypotheses of Proposition 2.3. Removing the extra interval gives

$$N_0^s(I_{\mathrm{th}}) \geq s_1 - N(I' \setminus I_{\mathrm{th}}) = s_1 - o(N),$$

and

$$\begin{aligned} N_d(I_{\mathrm{th}}) &\geq s_1 + s_2 + 2p - N(I' \setminus I_{\mathrm{th}}) \\ &\geq s_1 + s_2 + p - o(N). \end{aligned}$$

Consequently, (9) yields

$$N_0^s(T, 2T) \geq (2 - R(\psi) - o(1))N(T, 2T).$$

Similarly, (10) yields

$$N_d(T, 2T) \geq \frac{3 - R(\psi) - o(1)}{2} N(T, 2T).$$

Substituting $R = 4/3$ proves (1) and (2); substituting (27) proves (3).

The source proof bounds the relative error by $O_\psi(\log L/L) = O_\psi(\log\log T/\log T)$ for either fixed window above. This gives the rate in Theorem 1.1.

For the cumulative statement, decompose $(0, T]$ into $(T/2, T], (T/4, T/2], \ldots$ and stop when the remaining low-height interval is negligible compared with $N(0, T)$. Apply the dyadic estimate on each piece. The Riemann–von Mangoldt asymptotic and the geometric decay of the dyadic masses preserve the stated $O(\log\log T/\log T)$ relative error. The containments $N_0^s \leq N_0^* \leq N_0 \leq N$ and $N_0^s \leq N^s \leq N_d \leq N$ give (4). For a fixed primitive Dirichlet character χ, replace ζ by $L(s, \chi)$ in the explicit formula. The gamma factor and the $O(1)$ conductor term change only lower-order errors, $|\chi(n)| \leq 1$ preserves the prime estimates, and the finitely many primes dividing the fixed conductor contribute only lower-order terms. If χ is complex, conjugation first carries a zero of $L(s, \chi)$ to one of $L(s, \overline{\chi})$; the functional equation then carries it back to the reflected zero $1-\overline{\rho}$ of the same $L(s,\chi)$. Thus the off-line pairing is unchanged. The matrix data and its constants are unchanged. This completes the proof of Theorem 1.1.

6. Why these data cannot give better constants

The transfer principle uses only a first moment, a second moment, and the sign restriction from functional-equation pairs. These facts alone cannot give better constants. For example, let M be divisible by six and take mutually orthogonal contribution directions with spectrum

$$1^{(2M/3)}, \qquad 2^{(M/6)},$$

where the superscripts denote multiplicities. Its trace and squared Hilbert–Schmidt norm are

$$\frac{2M}{3}+2\frac{M}{6}=M, \qquad \frac{2M}{3}+4\frac{M}{6}=\frac{4M}{3}.$$

Interpreting the first block as simple on-line contributions and the second as on-line double contributions gives simple count $2M/3$ and distinct count $2M/3+M/6=5M/6$. Thus both constants are attained by the data visible to Proposition 2.3.

The theorem gives lower bounds; it does not say that the remaining third lies off the line. A better constant needs more information, such as higher moments outside the bandwidth-one prime-side range. Rearranging the same trace and energy data cannot improve it.

7. Earlier work

A lot of this history is already covered in the source. Here are the results used in the proof or most closely related to it.

Weil's explicit formula gives the Hermitian form [17]. Yoshida studied when it is positive for small Fourier support [19], and Bombieri related its positive and negative directions to off-line zeros [5]. Montgomery's pair-correlation method gave a conditional route to two thirds [12]; his ICM account gives the optimized Montgomery–Taylor constant [13]. Conrey, Ghosh, and Gonek supplied the multiplicity inequality behind the five-sixths value [9].

Aryan and Baluyot–Goldston–Suriajaya–Turnage-Butterbaugh proved the unconditional estimates used on the zero and prime sides [3, 4]. Recent narrow-box work showed why proving positivity term by term is difficult [10, 11]. Claude's key move is to charge off-line pairs by inertia, then combine rank, trace, and Hilbert–Schmidt energy. Earlier unconditional bounds were 5/12 for simple critical-line zeros [16] and 0.6603 for distinct zeros [18]. These are comparisons, not steps in this proof.

Two results about families of Dirichlet L-functions look similar but make different assumptions. Under GRH, Özlük proved an averaged 11/12 simple-zero proportion for Dirichlet L-functions [15], while Chandee, Lee, Liu, and Radziwiłł gave a 91% result for primitive-character families [7]. Those theorems average over characters and assume GRH. Claude's statement is unconditional and applies to each fixed primitive character. The family results are useful context, but they do not imply it.

8. Sources and Lean evidence

The mathematical theorem, the analytic estimates, and the decisive use of Hermitian inertia are Claude's result, communicated by Anthropic in August 2026 [8, 1]. Bombieri had earlier related the inertia of finite truncations of Weil's form to off-line zero pairs [5]. This note states the zero-counting step as Proposition 2.3 and spells out the join between the number theory and the linear algebra. The source manuscript credits Jarred Sumner with posing and guiding the investigation, Ralph Furman and Levent Alpöge with independent re-derivation, context, and responsibility for communication, and Eric Easley with orchestrating the Lean formalization [8].

The accompanying Lean 4 repository [2] is at tag `v1.0` and commit

`3635e74826a4c1fcece7d1cd2b6fa75e43a00510`.

The Lean comparator states the zeta-side 2/3 and 5/6 results in both dyadic and cumulative form. It also states the Montgomery–Taylor improvements and the dyadic versions for fixed primitive characters.

The frozen repository's `AUDIT.md` records successful builds and no `sorry` outside the deliberately incomplete challenge files. Its `#print axioms` reports give precisely

$$\{\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}\}.$$

These claims come from the repository's audit. The Lean toolchain was not rerun for this note, so this is not a new formal certificate.

Assistance disclosure

OpenAI Codex assisted with checking formulas, citations, and typesetting. The theorem and main proof idea are due to Claude/Anthropic as attributed above. The human preparer remains responsible for final verification of the manuscript and for any errors.

Appendix A. Exact counting definitions and Lean statements

This appendix fixes exactly what each count means. The phrase "two thirds of the zeros" otherwise hides three choices: the height interval, multiplicity, and the critical-line condition.

Set

$$\begin{aligned}\mathcal{Z}(T_1,T_2)=\{\rho\in\mathbb{C}:\zeta(\rho)=0,\ 0<\Re\rho<1,\\ T_1<\Im\rho\le T_2\},\qquad m(\rho)=\operatorname{ord}_\rho\zeta.\end{aligned}$$

The comparator's six zeta-side names and their mathematical values are

Lean identifier	Mathematical object
`Ncount`	$\sum_{\rho\in\mathcal{Z}(T_1,T_2)} m(\rho)$
`Ndist`	$\#\mathcal{Z}(T_1,T_2)$
`N0`	$\sum_{\substack{\rho\in\mathcal{Z}(T_1,T_2)\\ \Re\rho=1/2}} m(\rho)$
`N0star`	$\#\{\rho\in\mathcal{Z}(T_1,T_2):\Re\rho=1/2\}$
`N0simple`	$\#\{\rho\in\mathcal{Z}(T_1,T_2):\Re\rho=1/2,\ m(\rho)=1\}$
`Nsimple`	$\#\{\rho\in\mathcal{Z}(T_1,T_2):m(\rho)=1\}$

In Lean, $m(\rho)$ is `analyticOrderAt` applied to `riemannZeta`, followed by `toNat`; the set cardinalities are `Set.ncard`, and the sums are finite sums `finsum`. The interval is exactly $(T_1,T_2]$ and uses positive ordinate, not absolute ordinate.

The two headline dyadic declarations have the literal quantifier shape

$$\begin{aligned}\forall\varepsilon>0\ \exists T_0\ \forall T\ge T_0:&\quad (2/3-\varepsilon)N(T,2T)\le N_0^s(T,2T),\\ \forall\varepsilon>0\ \exists T_0\ \forall T\ge T_0:&\quad (5/6-\varepsilon)N(T,2T)\le N_d(T,2T).\end{aligned}$$

Their cumulative declarations replace $(T,2T]$ by $(0,T]$. The Montgomery–Taylor declarations replace 2/3 and 5/6 by $2-c_{\mathrm{MT}}^{-1}$ and $(3-c_{\mathrm{MT}}^{-1})/2$, without changing the quantifiers.

For a Dirichlet character χ modulo q, the comparator makes the same definitions with Mathlib's analytic continuation `DirichletCharacter.LFunction chi`. Its theorem declarations quantify over q with $q>1$ and over a primitive χ, then use the same ε,T_0,T order. Thus "fixed primitive character" means that T_0 may depend on χ but the two constants do not. These comparator declarations are dyadic. The cumulative fixed-character clause comes from Theorem 1.1 and dyadic summation, not from a cumulative Dirichlet declaration in this comparator topic.

All identifiers in this appendix come from the Mathlib-only trusted layer `comparator/ChallengeDeps.lean` at the frozen commit. These are theorem statements, not a new local Lean build.

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